

Mohammad Afzal

Lead Cloud Engineer | Senior DevOps Engineer | Product Development Engineer
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Summary

Lead Cloud Engineer at Technumen with 9+ years of experience in designing, building, and maintaining scalable, cloud-native systems. Specialized in **AWS infrastructure (EC2, Lambda, S3, RDS)**, **containerization (Docker, Kubernetes)**, and **CI/CD automation** using tools like **GitHub Actions, Jenkins, and GitOps workflows**. Strong backend development background in **Java, Kotlin, and Python**, enabling seamless integration between application development and deployment pipelines. Proficient in **Infrastructure as Code (Terraform)**, **observability tooling** (Prometheus, Grafana, ELK), and **DevSecOps best practices**. Experienced in enabling ML Ops using **MLflow, SageMaker, and Kedro**, supporting real-time model deployment and lifecycle automation. Adept at leading cross-functional teams, optimizing build and release processes, and delivering high-availability systems for product and service-based organizations. Active contributor on Stack Overflow (8,000+ reputation), with a passion for clean architecture, automation, and developer enablement.

Technical Proficiencies

- **Cloud Platforms:** AWS (EC2, Lambda, S3, RDS, EKS, CloudWatch, Step Functions), GCP (basic), Serverless Framework
- **Infrastructure as Code (IaC):** Terraform, AWS CloudFormation (basic), Kubernetes Manifests, Helm Charts
- **Containerization & Orchestration:** Docker, Kubernetes, Docker Compose, Helm, Amazon EKS
- **CI/CD & Automation:** GitHub Actions, Jenkins, GitLab CI/CD, GitOps workflows, Shell scripting
- **Programming & Scripting:** Python, Kotlin, Java 17, Bash, JavaScript (basic)
- **DevOps & Monitoring Tooling:** Prometheus, Grafana, ELK Stack, CloudWatch, Fluentd, Nginx, Kafka
- **Source Control & Dev Tools:** Git, GitHub, Bitbucket, GitLab, Linux CLI, VS Code, IntelliJ, Postman
- **ML Ops & Data Pipelines:** MLflow, SageMaker, Kedro, Pandas, JupyterLab, Scikit-learn
- **Testing & Quality Automation:** JUnit, PyTest, Cucumber, Postman, Selenium (basic), BDD frameworks
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, Snowflake (basic), S3

Experience

Lead Cloud Engineer, Liberty Mutual (via Technumen) | Mar 2024 – Jul 2025

- Oversaw the integration of advanced cloud services, including serverless computing models like AWS Lambda, to optimize operational efficiency and reduce costs.
- Led the development of scalable microservices using Java and Spring Boot, integrating Kafka for efficient message processing and AWS for robust cloud infrastructure.
- Streamlined deployment processes with Docker and Kubernetes, implementing CI/CD pipelines using **Jenkins** to ensure rapid, reliable delivery.
- Developed dynamic web interfaces using **ReactJS**, ensuring seamless integration with backend systems to enhance user experience and system performance.

Product Development Engineer, Calibo | Oct 2020 – Jan 2024

- Lead the development of scalable web applications in a microservice architecture using Java, Spring Boot, and Hibernate.
- Collaborate with cross-functional teams to understand requirements and deliver high-quality features to production.
- Architect and implement DevOps practices, leveraging Docker, Kubernetes, and Jenkins for continuous integration and deployment.
- Utilize **AWS** services such as **EC2, RDS, and Sagemaker** to build robust and scalable cloud solutions.

- Work closely with stakeholders to gather feedback, optimize application performance, and ensure smooth product delivery.

Senior Associate, Nagarro | Mar 2019 – Oct 2020

- Developed enterprise applications using a **hybrid microservices stack**—**Node.js** for lightweight APIs and **Kotlin and Java with Spring Boot** for core business logic.
- Led a small team of engineers in designing, developing, and maintaining microservices, ensuring high scalability and modularity.
- Led the digital transformation of project development by developing web applications using **Kotlin + Spring Boot, RabbitMQ, Angular 8, and Postgresql**.
- Actively participated in agile development processes, including sprint planning, daily stand-ups, and retrospective meetings.

Senior Software Engineer, Enbraun | Jul 2015 – May 2019

- **Core contributor and technical lead** on the development of *eResource Scheduler*, a multi-user resource management platform built from the ground up for scalability, availability, and customization on **AWS (EC2, RDS, S3, ALB)**.
- Designed and developed **RESTful services** and dynamic frontend modules using **JavaEE, AngularJS, and Bootstrap**, forming the backbone of the platform's booking, timesheet, and reporting features.
- Mentored junior developers and facilitated code reviews, fostering a culture of quality and hands-on learning within the team.
- Played a core role in the platform's **first public release after 3 years of active development**, contributing across all phases—from designing architecture and building features to testing, optimization, and deployment of a **highly available and scalable system**.

Lead Cloud Engineer, Technumen - Client: Liberty Mutual | Mar 2024 - Jul 2025

- Built and deployed a scalable **Spring Boot SDK** for AI-driven insurance inference; integrated with **Sapient** and third-party clients.
- Automated **release notifications** to internal and external stakeholders via **Slack, email, and Microsoft Graph API**.
- Managed AWS infrastructure (**EKS, Lambda, S3, Step Functions**) using **Terraform, Docker, and Kubernetes**.
- Designed CI/CD pipelines with **GitHub Actions**, enabling secure, multi-stage deployments and code scanning.
- Integrated monitoring and alerting with **Datadog, Prometheus, and Grafana** for production reliability.
- Migrated codebases to **GitHub**, implemented GitOps practices, and mentored team on **Kotlin** and infra best practices.

Product Development Engineer, Calibo | 2020 - 2023

- Work closely with stakeholders to gather feedback, optimize application performance, and ensure smooth product delivery.
- Built and deployed containerized microservices using **Kotlin, Spring Boot, Docker, and Kubernetes (EKS)**.
- Created and maintained Helm charts for service deployment and environment-specific configurations.
- Managed cloud infrastructure using **AWS (EC2, RDS, S3, SageMaker)** and **Terraform**, enabling repeatable, scalable provisioning.
- Developed and maintained **Jenkins pipelines** for CI/CD across multiple environments, integrating testing and code quality checks.
- Monitored system health and performance using **New Relic, CloudWatch**, and centralized logging with **ELK Stack**.
- Contributed to **ML Ops pipelines** using **MLflow, Kedro, and SageMaker** for model deployment and monitoring.
- Collaborated with teams to implement DevOps best practices, automate deployments, and troubleshoot production issues efficiently.

Senior Associate, Nagarro | 2019 - 2020

- Developed microservices using **Spring Boot**, **Kotlin**, and **RabbitMQ**, deployed on containerized infrastructure.
- Built and maintained **CI/CD pipelines** using **Jenkins** and **Docker**, streamlining testing and release workflows.
- Collaborated on infrastructure setup with **Kubernetes**, and implemented logging/monitoring with **ELK Stack** and **CloudWatch**.
- Contributed to DevOps transformation by automating deployments and improving code quality through static analysis and test coverage.
- Participated in agile ceremonies and cross-team collaboration to deliver scalable, secure features aligned with client needs.

Senior Software Engineer, Enbraun | 2015 - 2019

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Freelancer | Before 2015

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Project Profile

OneRule SDK – AI Inference Platform for Insurance Claims

Enterprise-grade SDK designed to simplify integration with Sapien's AI inference engine for insurance claim processing. The SDK abstracted data preparation and infrastructure complexities, providing a lightweight, plug-and-play middleware solution for third-party systems.

Key Contributions:

- Designed and developed the core SDK using **Java 17**, **Spring Boot**, and **Kotlin**, enabling low-latency access to AI-driven inference APIs.
- Containerized services with **Docker** and deployed them on **AWS EKS**, with configuration management using **Helm** charts.
- Managed infrastructure using **Terraform**, provisioning AWS components like **Lambda**, **S3**, **Step Functions**, and **CloudWatch**.
- Implemented CI/CD pipelines using **GitHub Actions** for automated builds, deployments, and environment promotion.
- Developed and published reusable **CloudForge Starter Templates** (e.g., Spring Boot Starter, Kubernetes-ready Helm charts, EKS deployment blueprints) to accelerate client onboarding.
- Integrated **Slack bots** and **Microsoft Graph API** to automate release announcements and internal communications.
- Introduced observability enhancements using **Datadog** for real-time monitoring and alerting.
- Migrated repositories from Bitbucket to GitHub with enforced GitOps workflows, code quality checks, and reviewer policies.

LAZSA – Product Platform-as-a-Service

An industry-first Product Platform-as-a-Service (pPaaS) built to streamline and accelerate digital product development. LAZSA enables enterprises to manage the complete lifecycle of digital solutions—from definition and design to deployment—within a unified, cloud-native environment.

Key Contributions:

- Led the development of core platform modules using **Java 8**, **Spring Boot**, and **Kotlin**, ensuring modularity, scalability, and ease of extension.
- Architected and deployed microservices using **Docker** and **Kubernetes**, with CI/CD pipelines built in **Jenkins** for automated testing and delivery.
- Integrated **AWS services** including **S3**, **Lambda**, **RDS**, and **SageMaker** for storage, computation, and ML model hosting.
- Implemented ML pipelines using **Kedro**, with experiment tracking and model lifecycle management via **MLflow**.
- Utilized **Kafka** for asynchronous, event-driven communication across distributed services.
- Developed frontend interfaces using **ReactJS**, and managed deployment environments via **Helm** charts and infrastructure as code.

- Established observability stack using **ELK**, **Grafana**, and **CloudWatch** for real-time monitoring, logging, and performance tuning.
- Collaborated with stakeholders to define platform strategy and ensure business-aligned technical decisions for speed, compliance, and scalability.

DiPEC (Digitalized Project Execution Cycle)

Web application created to digitalize project development, utilized by engineers of Siemens. The project involved team leadership and development from the initial stages. The application was developed in a microservices architecture, ensuring scalability.

Link : <https://www.dipec.siemens.co.in>

Key Contributions:

- **Led the end-to-end development** of critical features from inception, working closely with Siemens stakeholders to translate engineering workflows into intuitive application modules.
- Designed and implemented services using **Spring Boot** and **Kafka** to enable scalable, event-driven communication between modules.
- Developed a dynamic and responsive frontend using **Angular 8** and **Material UI**, integrated seamlessly with **PostgreSQL** for reliable data persistence.
- Took ownership of code quality, reviews, and sprint delivery, ensuring adherence to agile processes and successful on-time releases.

DiPEC (Digitalized Project Execution)

Web application created to digitalize project development, utilized by engineers of Siemens. The project involved team leadership and development from the initial stages. The application was developed in a microservices architecture, ensuring scalability.

Key Contributions:

- Built core services using **Spring Boot**, **Kotlin**, and **PostgreSQL** with **Kafka** for event-driven communication.
- Developed frontend in **Angular 8** with **Material UI**, ensuring responsive UX.
- Deployed services using **Docker** and **Kubernetes**; implemented logging via **ELK Stack** and **CloudWatch**.
- Collaborated with cross-functional teams to align delivery with agile sprints and performance targets.

eResource Scheduler

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ExamChi – Online Exam Practice Platform

A cloud-based exam preparation tool designed to deliver dynamic quizzes, performance tracking, and practice sessions for students and professionals. Built as a full-stack application to demonstrate scalable architecture, real-time user interaction, and cloud-native deployment.

Link : <https://examchi.afzalex.com/>

Key Contributions:

- Designed and developed the platform using **Node.js** (Express) for backend services, **ReactJS** for the frontend, and **MongoDB** for schema-flexible data storage.
- Deployed on **AWS** with a focus on performance and availability, leveraging services like EC2 and S3 for hosting and storage.

Abho Costing Sheet

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Education Detail

Master of Science in Data Science from Liverpool John Moores University (Distance)

Post Graduate Diploma in Data Science from IIITB Bangalore (Distance)

Bachelor of Technology in Computer Science from Uttarakhand Technical University

Higher Secondary School from Modern School Rishikesh (CISCE)

High School from Modern School Rishikesh (CISCE)

